

PRITOM MAJUMDER

B.Sc. in Materials and Metallurgical Engineering

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

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[LinkedIn](#) | [Research Portfolio](#) | [Undergraduate Thesis \(PDF\)](#)

RESEARCH INTERESTS

- **Functional & Piezoelectric Ceramics:** Lead-free ferroelectrics (KNN-based systems), solid-state synthesis, dielectric and piezoelectric response, and low-temperature sintering routes.
- **Oxide Semiconductor Thin Films:** Doping-driven control of structural, optical, magnetic and electronic behaviour in transparent conducting oxides.
- **Photonic & Optoelectronic Materials:** Linking structure-property understanding to working devices: lasers, LEDs, photovoltaics and magnetic data storage.

EDUCATION

B.Sc. in Materials and Metallurgical Engineering

2021 – June 2026

Bangladesh University of Engineering and Technology (BUET), Dhaka

CGPA: 3.64 / 4.00 Merit position: 15 of 59 in the graduating class

Relevant coursework: Advanced Ceramics; Photonics and Magnetic Materials; Electrical, Optical and Magnetic Properties of Materials; Nanostructured Materials & Thin Films; Characterisation of Materials; Crystallography and Structure of Materials; Phase Diagrams & Transformations; Mechanical Behaviour of Materials; Surface Engineering & Tribology; Corrosion & Degradation of Metals and Alloys.

Higher Secondary Certificate (Science)

2020

Notre Dame College, Dhaka GPA: 5.00 / 5.00 Board: Dhaka

PUBLICATIONS

In preparation

1. **Pritom Majumder**, Md. Atik Faisal, Mehrab Rahman, Anabil Dey, and Md. Azizar Rahman. “Structural, Optical, Magnetic, and Electronic Properties of Hydrothermally Grown Cu-Doped SnO₂ Thin Films: A Combined Experimental and First-Principles (DFT+U) Study.” Manuscript under review, 2026. **(First author)**

In preparation

2. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. “Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties — A Review.” Manuscript in preparation. **(First author)**

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher · Department of MME, BUET

2025 – 2026

“Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics” Supervisor: Prof. Dr. Ahmed Sharif

- Synthesized lead-free K_{0.5}Na_{0.5}NbO₃ (KNN) ceramics modified with Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O₃ (BCHfT) by the conventional solid-state route, as an environmentally safer alternative to lead-based PZT.
- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient d₃₃.
- Identified KNN-1 wt% BCHfT as the optimum composition, giving the best balance of dielectric response and low loss together with the highest unpoled d₃₃ measured in the series (15.5 pC/N).

SELECTED PROJECTS

Millsorp - Capstone Design Project · Team of 6, Department of MME, BUET

2025

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated hematite-rich iron-oxide adsorbent for heavy-metal wastewater treatment.
- Designed a low-cost bench-scale filtration column achieving up to 95.8% removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV-Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR and the Bangladesh Atomic Energy Commission, receiving positive technical feedback.

3D PotterBot - CAD Modelling & Finite Element Analysis · SolidWorks

2023

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118-part assembly; produced photorealistic renders, an exploded-view motion study, and a static FEA of the vertical support rod (≈15 mm peak deflection, strain and von Mises stress fields).

INDUSTRIAL & LABORATORY TRAINING

Bangladesh Council of Scientific and Industrial Research (BCSIR) · Dhaka

2026

- Performed the characterization for my undergraduate thesis on BCSIR instruments: XRD for phase analysis, SEM for grain structure, and temperature- and frequency-dependent dielectric measurement on a Wayne Kerr impedance analyser with a 1020 dielectric material fixture.

Abul Khair Steel Melting Limited (AKSML) · Chittagong

March 2025

- Studied electric arc furnace, ladle refining furnace and continuous casting operations, and rolling-mill practice using Quenching & Tempering Bar (QTB) technology for structural steel.
- Observed quality-control workflows spanning ICP-OES, XRF, handheld XRF and wet chemical analysis, together with process automation and real-time monitoring.

Bangladesh Industrial Technical Assistance Centre (BITAC) · Dhaka

March 2025

- Hands-on training in precision machining (lathe, milling, CNC), TIG/arc/spot welding, heat treatment, foundry operations, CAD-CAM and industrial automation.

TECHNICAL SKILLS

- Synthesis & Processing:** Solid-state ceramic synthesis; hydrothermal synthesis; sol-gel routes; sintering, uniaxial pressing and poling; solution casting.
- Structural & Microstructural:** X-ray diffraction (XRD, X'Pert HighScore); FE-SEM and SEM; TEM including HRTEM, SAED and EDS; optical metallography; ImageJ grain analysis.
- Electrical & Optical:** Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement; d_{33} meter; cyclic voltammetry; I-V and Mott-Schottky analysis; UV-Vis spectroscopy; photoluminescence; FTIR.
- Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis.
- Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator.

AWARDS & SCHOLARSHIPS

- University Merit Scholarship**, Bangladesh University of Engineering and Technology (July 2022)
- University Stipend Scholarship**, Bangladesh University of Engineering and Technology (July 2021 and January 2022)
- Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training, 2025.
- Certificate of Perfect Attendance**, Notre Dame College, Dhaka - awarded for full attendance across the two-year HSC programme.

LANGUAGE PROFICIENCY

- English:** IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5), 2026.
- Bangla:** Native.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- Private Tutor**, intermediate-level students, 2022-2026 :- delivered regular one-to-one academic tuition alongside full-time undergraduate study.
- Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer.
- Class Representative**, Department of MME, BUET, 2021-22 :- elected by classmates for both semesters of the first year.
- Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present.

REFERENCES

Prof. Dr. Ahmed Sharif

Professor, PhD

Department of Materials and Metallurgical Engineering, BUET

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Undergraduate Thesis Supervisor

Dr. Md. Moniruzzaman

Professor

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